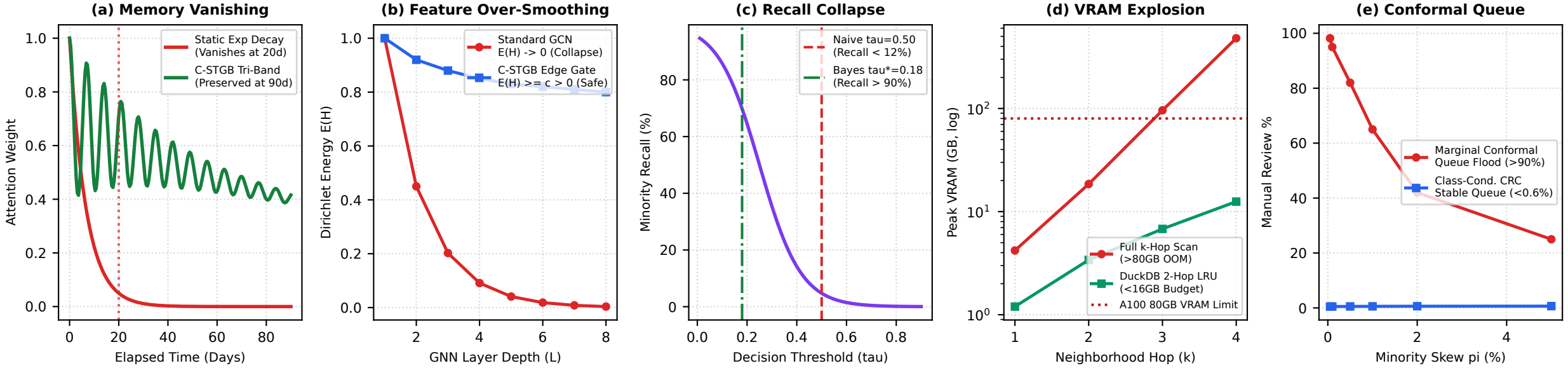
(a) Memory Vanishing
Static Exp Decay
(Vanishes at 20d)
C-STGB Tri-Band
(Preserved at 90d)
Attention Weight
Elapsed Time (Days)
(b) Feature Over-Smoothing
Standard GCN
E(H) -> 0 (Collapse)
C-STGB Edge Gate
E(H) >= c > 0 (Safe)
Dirichlet Energy E(H)
GNN Layer Depth (L)
(c) Recall Collapse
Naive tau=0.50
(Recall < 12%)
Bayes tau*=0.18
(Recall > 90%)
Minority Recall (%)
Decision Threshold (tau)
(d) VRAM Explosion
Full k-Hop Scan
(>80GB OOM)
DuckDB 2-Hop LRU
(<16GB Budget)
A100 80GB VRAM Limit
Peak VRAM (GB, log)
Neighborhood Hop (k)
(e) Conformal Queue
Marginal Conformal
Queue Flood (>90%)
Class-Cond. CRC
Stable Queue (<0.6%)
Manual Review %
Minority Skew pi (%)